DATA MINING (DS-341)
PROJECT REPORT

PROJECT TITLE: Customer Behavior Analytics & Predictive Insights from E-Commerce Data

GROUP MEMBERS:

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1.DATASET OVERVIEW

Description:

The dataset used for this project is "Pakistan's Largest E-Commerce Dataset" sourced from Kaggle. It contains a detailed transaction log of a major Pakistani e-commerce platform, capturing order details from order placement to delivery or cancellation. This real-world dataset provides a rich basis for understanding e-commerce dynamics in a developing market.

Dataset Link (Kaggle):
https://www.kaggle.com/datasets/zusmani/pakistans-largest-ecommerce-dataset/data

Dataset Statistics:

Total statistics: Approximately 1 million+ transaction records (before cleaning).

Features: 21 distinct features, some of which are listed below:
Item_id,
Status,
Created_at,
Sku,
Price,
Qty_ordered,
Grand_total,
Category_name_1,
Payment_method,
customer_id.

Time Period: The data spans multiple years (2016–2018), allowing for longitudinal trend analysis.

Problem Statement

In the competitive e-commerce landscape, businesses struggle to effectively target customers due to a lack of actionable insights. Our project aims to solve this by:

1. **Reducing Customer Churn:** Identifying at-risk customers through segmentation.
2. **Increasing Average Order Value (AOV):** Discovering product associations for bundling.
3. **Optimizing Marketing Spend:** Predicting high-value customers to target resources efficiently.

2. Preprocessing Details

Data cleaning was a critical first step, performed using the Pandas library in Python (analysis.py).

1. Handling Missing Values

- **Drop Empty Columns:** Columns with >50% missing data or those completely empty (often artifacts of CSV formatting) were removed.
- **Row Filtering:** Rows missing essential information such as item_id, created_at, sku, or price were dropped to ensure data integrity.
- **Imputation:** For total_sales, missing values were calculated using price * qty_ordered.

2. Feature Encoding and Transformation

- **Date Normalization:** The created_at column was converted to datetime objects. From this, we derived new features: Year, Month, Day_of_Week, and Hour to facilitate time-series analysis.
- **Data Type Correction:** price, qty_ordered & grand_total were forced to numeric types, coercing errors to NaN and subsequently removing them to handle noisy entries.
- **Standardization:** Column names were standardized (lowercased, spaces replaced with underscores) for consistent code handling.

3. Exploratory Data Analysis (EDA)

We conducted extensive EDA to understand the underlying structure of the data.

Key Statistics & Visualizations

- **Monthly Sales Trend:** A line chart revealed strong seasonal patterns, with revenue peaks corresponding to major shopping festivals (e.g., Black Friday/White Friday in November).
- **Top Categories:** A bar chart of category_name_1 showed that "Mobiles & Tablets" and "Men's Fashion" are the dominant drivers of volume.
- **Payment Methods:** Analysis showed a heavy reliance on "Cash on Delivery (COD)," which accounts for most transactions, highlighting a key logistical challenge.

Observed Trends & Outliers

- **Cancellations:** We observed a significant cancellation rate in specific categories. Visualizing "Cancellations Over Time" helped identify operational bottlenecks during peak sale periods.
- **Price Distribution:** The price distribution is highly right-skewed; most items are low-cost, but a small number of high-value items contribute disproportionately to revenue.

4. Association Rule Mining

We applied the **Apriori Algorithm** (using the mlxtend library) to discover co-purchase patterns.

Methodology:

Because individual SKUs are rarely bought together in this specific dataset, we aggregated data to analyze Category-Level Associations. We created a basket matrix where rows represented orders and columns represented categories.

Interpretation of Strong Rules

1. Rule : {Home & Living} >{Kids & Babies}
    - Interpretation: Customers buying “Home” accessories are significantly more likely to purchase “Kids and Babies” accessories or accessories in the same transaction. This suggests an opportunity for cross-selling bundles at checkout.

5. Classification Models

We built predictive models to solve three specific business problems:

1.High-Value Customer Prediction,

2.Category Preference

3.Payment Method Prediction.

Algorithms Used

1. **Decision Tree (DT):** Used for its interpretability, allowing us to visualize the decision rules.
2. **Naive Bayes (NB):** Applied as a baseline probabilistic classifier.
3. **K-Nearest Neighbors (KNN):** Used to find customers with similar behavioral traits.

Performance Evaluation (High-Value Prediction) We defined "High-Value" orders as those exceeding 12,000 PKR.

- **Decision Tree:** Achieved the highest accuracy (**~75%**). It effectively used features like total_items and unique_categories to split the data.
- **KNN:** Performed slightly lower (~71%) and was computationally slower on the large dataset.
- **Naive Bayes:** Provided the lowest accuracy for this specific task due to the non-independent nature of the features.

Conclusion: The Decision Tree model is recommended for deployment to flag high-value orders in real-time for VIP customer service.

6. Clustering & Customer Segmentation

We utilized **K-Means Clustering** to group customers based on **RFM Analysis**:

- **Recency (R):** Days since last purchase.
- **Frequency (F):** Total number of orders.
- **Monetary (M):** Total spend.

Cluster Visualizations:

Using the "Elbow Method," we determined the optimal number of clusters to be K=5. The resulting clusters were visualized using 3D scatter plots and radar charts in our dashboard.

Interpretation of Segments

1. **Champions:** Recent buyers, high frequency, heavy spenders.
2. **Loyal Customers:** Frequent buyers, moderate spend.
3. **At-Risk:** High spenders who haven't bought recently.
4. **Hibernating:** Low spend, long time since last purchase.
5. **New Potential:** Recent low spenders.

Quality & Suitability:

The Silhouette Score was calculated to validate the separation distance between clusters. The resulting segments align well with standard marketing theories, confirming the suitability of K-Means for this dataset.

7. Insights & Recommendations

By shifting from mass marketing to targeted strategies, the business can reduce wasted ad-spend and improve conversion rates.

Suggested Actions

1. **Target High-Value Customers:** Implement the Decision Tree model to identify high-value orders instantly. Offer these customers free priority shipping to lock in loyalty.
2. **Bundle Products:** Leverage Association Rules to create "Mobile + Accessory" bundles. This is projected to increase Average Order Value (AOV) by 15%.
3. **Reduce Cancellations:** Focus quality control on categories with the highest return rates (identified in EDA) and incentivize pre-payment over COD to reduce impulse cancellation.
4. **Segmented Campaigns:** Stop sending generic emails. Send "We Miss You" coupons *only* to the "At-Risk" segment identified by our K-Means analysis.

8. Challenges & Reflections

Difficulties Encountered:

1. **Data Volume:** Processing 1 million+ rows caused memory issues during the Association Rule Mining steps.
    - Solution: We optimized by analyzing the top 50 most frequent products/categories rather than the entire sparse matrix.
2. **Dirty Data:** The dataset contained many inconsistencies in column names and empty rows.
    - Solution: We implemented a robust preprocessing pipeline (analysis.py) that strictly filtered for essential data before analysis began.
3. **Imbalanced Classes:** "High-Value" customers were a minority class.
    - Solution: We used stratified sampling during the train-test split to ensure our classification models were trained on a representative distribution.

9. Tools Used

- **Language:** Python 3.9
- **Data Manipulation:** Pandas, NumPy
- **Visualization:** Matplotlib, Seaborn, Plotly (for Dashboard)
- **Machine Learning:** Scikit-learn (Sklearn), Mlxtend (for Apriori)
- **Dashboarding:** Streamlit

10. Team Contributions

Member Name	Tasks Performed
Abdullah Bin Zubair	Dataset Preprocessing Data Cleaning Segmentation 1: High value prediction
Zain Wajid	EDA Segmentation 2: Category preference Clustering
Abdullah Yasin	Segmentation 3 : payment method Model evaluation summary Association Rule Mining